

Long-term changes in soil salinity as influenced by subsoil thickness in a reclaimed coal mine in east-central Alberta

S. O. Olatuyi and L. A. Leskiw¹

*Paragon Soil and Environmental Consulting, 14805-119 Avenue, Edmonton, Alberta, Canada T5L 2N9.
Received 28 April 2014, accepted 26 August 2014. Published on the web 31 August 2014.*

Olatuyi, S. O. and Leskiw, L. A. 2014. **Long-term changes in soil salinity as influenced by subsoil thickness in a reclaimed coal mine in east-central Alberta.** *Can. J. Soil Sci.* **94**: 605–620. Elevated salinity and sodicity are major challenges to reclamation of surface-mined coal sites in the Alberta Plains region. Research plots were established in 1981 at the Battle River Coal Mine near Forestburg, AB, to determine an optimum depth of subsoil replacement over sodic mine spoil required to sustain agricultural capability. Subsoil thickness was varied at 0, 25, 65, 135, 165 and 335 cm, overlain with 15 cm topsoil. The plots were monitored annually from 1982 to 1986 and were seeded to forage from 1987. Plots were resampled in 2013 to examine long-term changes in soil quality by comparing the results with the historical means for the 1980s. Key soil parameters measured are pH, electrical conductivity (EC), total dissolved salts (TDS), sodium adsorption ratio (SAR) and soluble Na. The soil quality of the root zone (0–40 cm) improved over time in all treatments as EC, TDS, SAR and concentration of Na decreased significantly between the 1980s and 2013. Amounts of EC and soluble Na consistently increased with depth, suggesting salt accumulation was predominantly by downward leaching, but with contribution by upward migration of Na from the underlying spoil. Relative to the native soils in 2013, the root zone quality ratings for EC and SAR in the topsoil/spoil treatment were better than in the shallow-bedrock Solonetzic soil. Ratings for the 25-cm subsoil treatment were also comparable with the local Chernozemic soil, but better than the fine till Solonetzic soil. This study demonstrates that a minimum of 25 cm subsoil and 15 cm topsoil are required for mitigating salinity and sodicity in these reconstructed soils. The resampling in 2013 demonstrates that long-term monitoring is essential to obtain a better understanding of reclamation outcomes.

Key words: Soil quality, subsoil, spoil, root zone, reclamation, sodicity, salinity

Olatuyi, S. O. et Leskiw, L. A. 2014. **Influence de l'épaisseur du sous-sol sur l'évolution à long terme de la salinité dans le sol restauré d'une mine de charbon du centre-est de l'Alberta.** *Can. J. Soil Sci.* **94**: 605–620. Une salinité et une sodicité élevées peuvent constituer un obstacle majeur à la restauration des mines de charbon exploitées à ciel ouvert dans la région des plaines de l'Alberta. En 1981, des parcelles expérimentales ont été aménagées à la mine de Battle River, près de Forestburg, en Alberta, en vue d'établir la profondeur optimale du sous-sol de remplacement sur des déblais sodiques pour en permettre une exploitation agricole soutenue. Le sous-sol avait 0, 25, 65, 135, 165 ou 335 cm d'épaisseur et était recouvert de 15 cm de sol de surface. Les parcelles ont été contrôlées chaque année, de 1982 à 1986, et ensimencées pour la production de fourrage en 1987. Elles ont fait l'objet d'un nouvel échantillonnage en 2013, ce qui a permis de préciser les changements à long terme survenus au niveau de la qualité du sol par comparaison avec les moyennes historiques des années 1980. Les principaux paramètres étudiés étaient le pH, la conductivité électrique (CE), la concentration totale de sels dissous (CSD), le ratio d'adsorption du sodium (RAS) et la concentration de Na soluble. La qualité du sol dans la zone racinaire (0 à 40 cm de profondeur) s'améliore avec le temps pour tous les traitements, la CE, la CSD, le RAS et la concentration de Na ayant diminué de façon significative entre les années 1980 et l'année 2013. La CE et la teneur en Na soluble augmentent constamment avec la profondeur, signe que le sel s'accumule surtout par lixiviation vers le bas, mais qu'il y a aussi migration du Na des déblais sous-jacents vers le haut. Comparativement aux sols naturels, en 2013, la qualité du sol de la zone racinaire relativement à la CE et au RAS était supérieure pour le traitement sol de surface/déblais que pour le solonetz peu profond sur substrat rocheux. Les cotes obtenues pour la couche de 25 cm de sous-sol étaient comparables à celles du tchernoziom local, mais dépassaient celles du solonetz à moraine fine. Cette étude montre qu'il faut au moins 25 cm de sous-sol et 15 cm de sol de surface pour compenser la salinité et la sodicité de ces sols reconstitués. L'échantillonnage de 2013 indique qu'il est essentiel d'effectuer une surveillance à long terme pour mieux comprendre les conséquences des travaux de restauration.

Mots clés: Qualité du sol, sous-sol, déblais, zone racinaire, restauration, sodicité, salinité

¹Corresponding author (e-mail: lleskiw@paragonsoil.com).

Abbreviations: BRSRP, Battle River Soil Reconstruction Project; EC, electrical conductivity; SAR, sodium adsorption ratio; TDS, total dissolved salts

Sodic mine spoil presents major technical problems in reclamation of surface-mined coal sites in east-central Alberta. Restoring and sustaining land capability and productivity to pre-mined levels in this dynamic environment is complex and challenging. The most popular method for ameliorating problems caused by sodic mine spoils involves the placement of good quality topsoil and subsoil above the mine spoil to provide a root zone with favourable chemical and physical properties. The benefits of topsoil placement over spoil have been documented in previous studies (Sandoval et al. 1973; United States Department of Agriculture 1977; Grandt 1978; Nielsen and Miller 1980). The depth of soil required for adequate reclamation depends on environmental factors, intended land use and the properties of the spoil and soil materials. Soil thickness requirements for saline and sodic spoils may need to be greater than for other spoils to allow for upward salt migration, settling and subsidence, surface erosion, uneven spreading of soil material and internal drainage restrictions (Power et al. 1978). Other studies revealed that optimum crop yields were obtained at soil depths with various combinations of topsoil, subsoil and suitable spoil, ranging from about 50 to 120 cm (Power et al. 1981; Barth and Martin 1984; Doll et al. 1984; Merrill et al. 1985; Oddie et al. 1989).

While investigating the optimum subsoil thickness for reclaiming a sodic mine spoil near Highvale, AB, Oddie et al. (1989) observed that replacement of two different subsoil depths as 55 cm and 95 cm, capped with 15 cm of topsoil, were sufficient to restore the post-mine productivity of barley (*Hordeum vulgare* L.) and alfalfa-smooth brome grass (*Medicago sativa* L., *Bromus inermis* Leyss). However, there was no significant difference between the two subsoil depths. Long-term effects of coversoil depth on plant community were monitored in a reclaimed coal mine in northwestern New Mexico by Buchanan et al. (2005). The study showed the greatest species diversity with coversoil depth between 25 and 65 cm. McGinnies and Nicholas (1983) also found a positive relationship between coversoil depth and root and herbage production of intermediate wheatgrass (*Agropyron intermedium*), but there was no additional benefit to root biomass with coversoil depth beyond 40 cm. The crop performance in these studies was largely attributed to subsoil thickness and soil water storage. However, there was little information quantifying the chemical attributes of the reconstructed soil and the relative effects of subsoil thickness on salinity and sodicity.

Surface-mined coal fields in the Alberta Plains are dominated by Solonchic soils where the bedrock of the coal-bearing formations commonly occurs close to the surface. The utilization of the native soils as reclamation materials is constrained by high salinity, adverse physical properties and very thin A horizon, which are further complicated by the climatic and hydrogeologic conditions typical of this region (Leskiw 1989). As such, reclama-

tion of lands with extensive areas of Solonchic soils is challenging, and salvage of suitable quantities of topsoil is often difficult. The subsoils are also shallow with undesirable characteristics resulting from the accumulation of clay and sodium in the Bnt horizon. During stripping of soil materials prior to mining, the sodium-enriched B horizon can be mixed with the calcium-enriched upper C horizon for replacement over the spoil during reclamation, thereby resulting in elevated salinity and sodicity of the reconstructed soil (Leskiw 1989; Moran et al. 1989).

Following soil placement, dissolved salts in the surface materials can be leached by both saturated and unsaturated percolation into and through the profile. Where rainfall is sufficient, saturated flow through the soil will occur, with the wetting front proceeding at a uniform rate through the soil, provided materials are uniform. Where surface fissures and cracks are common, unsaturated flow can occur rapidly, thereby generating non-equilibrium water movement (Beven and Germann 1982; Kung 1990a, b). In 70 sites that had been reclaimed for more than 10 yr in North Dakota, Halvorson (1985) found increases in Na concentration to an average depth of 38 cm beneath the soil-spoil interface; indicating that Na had been leached out of the upper profile. Under dryland agricultural conditions in the Alberta Plains region, Macyk et al. (1995) observed that salts were leached or redistributed downward over time in the reclaimed soil profiles.

Upward migration of salts from sodic mine spoil into the overlying soil materials is also a major concern in reclamation of post-mining lands in the Great Plains Region (Merrill et al. 1983). This salt movement pattern has been attributed to various transport mechanisms, including capillarity, diffusion and evaporation. Upward salt movement by capillary rise, whereby water and dissolved solutes rise through the pores into the rooting zone is dominant in soils with shallow water tables. Moran et al. (1987) also reported the impacts of hydrologic process on salt migration for reclaimed soils in western Canada. In central Alberta, Moran et al. (1987) found that in reconstructed soils where the water table occurred within 1 m of the soil surface, soils could be expected to become saline and sodic through capillary rise of dissolved salts. In cases where water tables were deeper, downward leaching of salts predominated.

Using laboratory column studies and application of laboratory results to field situations in North Dakota, Merrill et al. (1983) found that diffusion and convection (advection) are important mechanisms of upward salt movement in reclaimed soils. For diffusion to be significant there must be sufficient concentration gradient at the soil-spoil interface. However, Leskiw (1989) indicated it is unlikely that Na could be carried farther than 10 to 15 cm upward into non-sodic materials in the Alberta Plains region. Upward migration of salt to a similar height has been reported in a number of studies

in the Northern Great Plains (Merrill et al. 1980; Barth 1983; Barth and Martin 1984) and in oil sands reclamation in north-eastern Alberta (Kessler et al. 2010). Upward migration of salt by evaporation is attributed to advective vapour flow in response to thermal and suction gradients (Parlange et al. 1998; Hillel 2004). Under moisture deficit conditions where rainfall events are followed by dry periods, advective vapour flow is an important mechanism for upward salt migration (Moran et al. 1987; Leskiw 1989). Depending on the prevailing site conditions, however, there could be a combination of any of the three mechanisms of upward salt migration. For instance, capillary rise and evapotranspiration of saline groundwater are two important pedologic processes of salt translocation in Solonchic soils (Fullerton and Pawluk 1987).

The ultimate goal of land reclamation in the Alberta Plains is to return disturbed lands to agricultural capability levels that are equivalent to pre-mined conditions. The Alberta Soil Quality Criteria (Alberta Agriculture 1987) is one of the standard methods for assessing the soil quality of reclaimed lands following disturbance and extraction of coal resources in Alberta (Cheel et al. 1994). The primary requirements for achieving this objective include the design of effective soil placement configurations for minimizing compaction and for mitigating salinity and sodicity in the reconstructed soil profiles. The suitability of reclamation materials is evaluated by assessing the soil quality for various chemical parameters that are critical to crop production and agricultural capability.

The present study was initiated in 1979 by the Reclamation Research Technical Advisory Committee (RRTAC) of the Alberta Land Conservation and Reclamation Council in collaboration with the coal industry. The study was designated as the Battle River Soil Reconstruction Project (BRSRP). The research framework involved the establishment of various field trials to monitor soil quality and crop performance on reclaimed lands that were previously disturbed by surface coal mining at the Battle River Coal Mine. The study was originally established to investigate four different experiments with the following objectives: (1) Subsoil Depth, to determine the optimum depth of subsoil replacement over sodic mine spoil required to sustain agricultural capability; (2) Torlea Soil, to assess the effects of various salvageable soil horizons and horizon blends on root zone quality of the Torlea Series, which are Solonchic soils with shallow bedrock; (3) Bottom Ash, to assess the potential utilization of bottom ash and gypsum as a capillary barrier to upward movement of salts from spoil; and (4) Slope Drainage, to determine the effects of slope position, spoil levelling and slope aspect on salt movement and crop productivity.

The research plots were monitored annually from 1982 to 1986 and were seeded to forage from 1987. The site was inspected by Paragon Soil and Environmental Consulting Inc. (Paragon) in 2012 and the plots were

found to be mostly intact. Hence, soil monitoring was conducted by Paragon in fall 2013. Of the four experiments initially established on the site, only the Subsoil Depth and Slope Drainage plots were sampled in 2013. However, this study reports the findings for the Subsoil Depth experiment. The objectives of the Subsoil Depth monitoring in 2013 were: (1) to determine the long-term effects of reclamation prescriptions on soil chemistry status, and to test if differences among subsoil treatments persist since the 1980s; (2) to verify whether the root zone quality improved, deteriorated or remained unchanged over the past 31 yr (1982 to 2013); and (3) to examine the soil quality ratings in the reclaimed soil relative to the local natural soils.

MATERIALS AND METHODS

Detailed descriptions of project setting, plot construction and other management practices pertinent to this study have been reported in a previous document (Leskiw 1989). Hence, only a brief description of the soil profile reconstruction is provided in the present study but with emphasis on the 2013 soil sampling program.

Site Location and Project Setting

The BRSRP site is located in east-central Alberta (lat. 52°27'20"–52°27'30"N, long. 112°7'53"–112°8'3"W), 12.5 km southwest of Forestburg. This region experiences a continental climate characterized by warm summers and cold winters. January is the coldest month with a mean temperature of -16°C and July is the warmest month with a mean temperature of 17°C (Fig. 1). The average climate data were obtained from the Provincial weather record for Forestburg (Forestburg AGCM, AgroClimatic Information Service, Alberta Agriculture and Rural Development). The total annual precipitation based on the recent 6-yr average (2008–2013) was 350 mm, much lower than 400 mm for the 30-yr normal (1981–2010). The 6-yr average for total annual evapotranspiration (estimated for grass) is 809 mm. Of the total annual precipitation, about 270 mm occurs as rainfall during the growing season (May to September) according to the 2008–2013 average (288 mm for the 30-yr normal), while the total evapotranspiration for this period is 624 mm. The growing season total rainfall recorded on the site between 1983 and 1986 ranged from 200 to 276 mm. The climate data indicated a moisture deficit at this region as the total evapotranspiration was about twice the amount of precipitation, though this difference is expected to be lower for evapotranspiration based on annual crops.

The study area lies within the Castor Plain Physiographic District, an area of morainal veneer and blanket overlying undulating bedrock of the lower Horseshoe Canyon Formation (Pettapiece 1986). This bedrock, which occurs above coal seams, is saline, sodic and high in clay, and is rated "Unsuitable" for reclamation (Alberta Agriculture 1987). The dominant soil subgroups

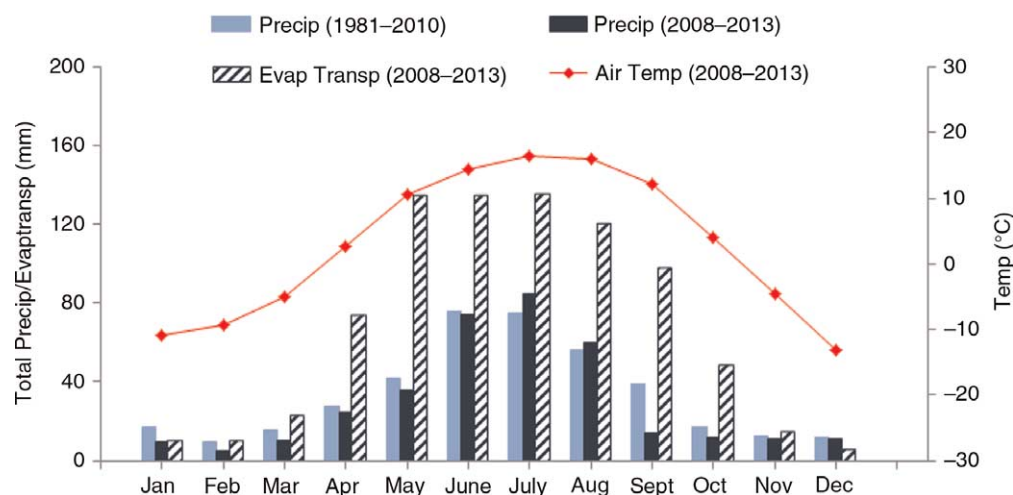
Long-Term Average of Monthly Climate Data
for Forestburg, AB

Fig. 1. Climate data based on recent 6-yr average (2008–2013) for precipitation (Precip), evapotranspiration (EvapTransp) and air temperature (temp), and a 30-yr normal (1981–2010) for precipitation. Data obtained from the Provincial weather record for Forestburg, AB. Forestburg AGCM, AgroClimatic Information Service, Alberta Agriculture and Rural Development.

comprise the Solonchic Dark Brown Chernozem and Dark Brown Solonchic Solonchic. The Chernozem soil is represented by the Flagstaff Series while the Solonchic Solonchic is dominated by the Halkirk Series in areas where the parent materials are glacial till. The Torlea Series occurs in areas where the residual materials for the Solonchic Solonchic are found at or near the surface (Well and Nikiforuk 1988). Arable soils in the region are used for dryland crop production, mainly wheat, barley and canola, while the non-arable lands are utilized for pasture and forage production.

Plot Construction and Research Monitoring

Plots were constructed on levelled spoil in 1980 using dozers and scrapers (Manalta Coal Ltd., Vesta Mine Machinery). Construction simulated the “take and put” technique of mine reclamation described by Grandt (1978). The plot layout was established as a randomized complete block design (RCBD) for the four experiments as shown in Fig. 2. The Subsoil Depth experimental units comprised three replicate blocks with two parallel blocks running north–south and the third block running east–west (Fig. 2). Each block was 151 m long and was demarcated into 12 treatment plots. The plot dimension was 8 m × 20 m, separated by 5 m of buffer zone. The subsoil thicknesses across treatment plots were 0, 25, 65, 135, 165 and 335 cm denoted as SUB-0, SUB-25, SUB-50, SUB-100, SUB-150 and SUB-300, respectively. Each treatment plot was overlain with 15-cm cover of topsoil (A horizon material). Due to variability in the actual depths of soil layers relative to the original design specifications following soil placement, the average value of actual subsoil thickness was reported for each treatment.

Following plot construction and soil placement, the soil profiles were sampled in 1981 to characterize each treatment plot for baseline properties. Soil samples were collected at various depths, air-dried and analyzed for pH (1:2 soil:water) and electrical conductivity (EC), sodium adsorption ratio (SAR) and soluble Na using the saturated paste extraction procedure (McKeague 1978). Soil analyses in the 1980s were performed in a commercial soil-testing laboratory by Norwest Soil Research Ltd., Edmonton, AB. The baseline and subsequent soil quality were rated for reclamation suitability according to the Alberta Soil Quality Criteria (Alberta Agriculture 1987). A summary of soil quality assessments of the materials used for reclamation is shown in Table 1.

The replicate blocks were seeded to two cropping systems including cereal rotation and forage production. Subsoil treatments and cropping systems were completely randomized within each block. Soil and crop husbandry, such as fertilizer application, seeding, weed and pest control, were carried out using conventional agronomic management practices (Leskiw 1989). Plots were monitored annually for various soil and crop parameters from 1982 to 1986. Following the initial 5-yr monitoring in the 1980s, all plots were seeded to forage in 1987 and remained intact without monitoring or agronomic activity. Due to a recent mine development nearby, a portion of the study site was covered by spoil material, burying some of the experimental units located at the western portion of the site (Fig. 2). This resulted in loss of one of the three experimental blocks and a couple of plots in another block for the Subsoil Depth experiment. Nonetheless, it was possible to obtain representative plots for each treatment in the

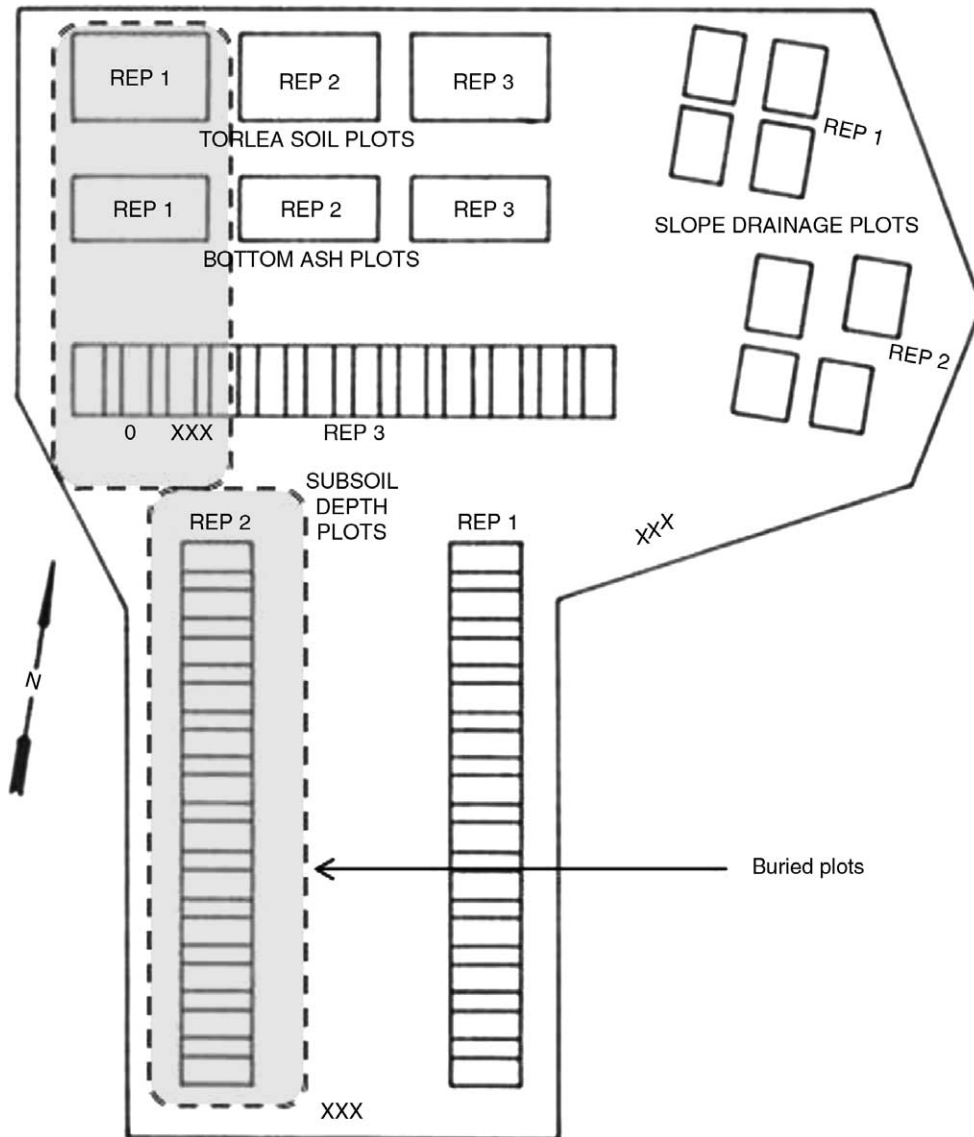


Fig. 2. Plot layout containing the original four experiments (Torlea Soil, Bottom Ash, Slope Drainage and Subsoil Depth). The figure also shows some buried plots on the west section of the site. Figure not drawn to scale. The geographical coordinates for plot location are lat. 52°27'20"–52°27'30"N, long. 112°7'53"–112°8'3"W.

remaining two blocks by combining the subplots originally seeded to cereal and forage in the 1980s.

Soil Monitoring in 2013

The site was inspected by Paragon Soil and Environmental Consulting (Paragon) in 2012 and was mostly intact under forage. Soil monitoring recommenced in fall 2013 and each treatment plot in the Subsoil Depth experiment was sampled to 120 cm depth at intervals of 0–15, 15–40, 40–100 and 100–120 cm. The sampled soil layers are denoted as topsoil (0–15 cm), upper subsoil (15–40 cm), mid subsoil (40–100 cm) and lower subsoil (100–120 cm). Soil samples in 2013 were analyzed by Maxxam Analytics Inc., Edmonton, AB, for pH (1:2

soil:0.01 M CaCl₂), EC, SAR and soluble Na using the saturated paste extraction method, the same as that in the 1980s (Miller and Curtin 2008; Hendershot et al. 2008). To determine amounts of salt loading in the soil layers, a mass balance approach was used to estimate total dissolved salts (TDS) in the solid phase based on the following equations:

$$\text{TDS (mg L}^{-1}\text{)} = \text{EC} \times 800 \quad (1)$$

$$\text{TDS (t ha}^{-1}\text{)} = \text{TDS (mg L}^{-1}\text{)} \times \text{saturation \%} \\ \times \text{bulk density} \times \text{depth} \quad (2)$$

where TDS (mg L⁻¹) is the total dissolved salts in solution, while EC (dS m⁻¹) was measured by the

Table 1. Soil quality assessment of reclamation materials used for the Subsoil Depth experiment

Material	Salvaged horizon	Texture	Rating ^z	Comments
Topsoil	Ah or Ap	Loam to clay loam	Fair	The major limitation was salinity (EC ranged from 2 to 4 dS m ⁻¹)
Subsoil	B and C plus underlying material (shallow till and weathered rock) with physical properties suitable for vegetative growth	Clay loam	Fair to Poor	High salinity and sodicity were major limitations (mean EC and SAR was 5.8 dS m ⁻¹ and 8.4, respectively)
Spoil	Bedrock materials of the Horseshoe Canyon Formation	Clay	Unsuitable	Excessive sodicity (SAR > 20) but low salinity (EC < 3.0 dS m ⁻¹)

^zSoil quality rating for reclamation suitability according to the Alberta Soil Quality Criteria for the Plains region (Alberta Agriculture 1987).

saturated paste procedure. The conversion factor of 800 was based on the value recommended for soils with EC > 5.0 dS m⁻¹ (Alberta Environment 2001). The salinity data were obtained from a saturated paste extract, bulk density (g cm⁻³) was measured by soil coring method, and depth (cm) is the thickness of each soil layer.

Statistical Analysis and Soil Quality Ratings

The vertical distribution and variations in soil chemistry status among treatments in 2013 were determined by two-way analysis of variance (ANOVA) of soil layer × treatment effect using the MIXED procedure in SAS software (SAS Institute Inc.). To generate a trend analysis of long-term changes in soil chemistry, the 2013 results were compared with the historical mean for the 1980s soil monitoring. Due to some missing datasets for 1985 and 1986 monitoring, the historical mean for the 1980s constitutes only the 1982–1984 data. Temporal variations in soil chemistry and treatment effects were tested by two-way ANOVA of year × treatment.

Though the plots were originally established as a randomized complete block design, there was a slight deviation in the statistical analysis due to loss of some experimental units buried by spoil on the west section of the site (Fig. 2). However, it was possible to obtain complete datasets and replicates for each treatment by combining both cereal and forage subplots in the 1980s and 2013. Therefore, the ANOVA model was based on an incomplete block design (Littell et al. 2006), which included replicate as a random factor with subsoil treatment and sampled soil layer or year of monitoring as fixed factors. The soil layer was treated as a repeated measure factor and the spatial power [SP(POW)] covariance structure was used for the repeated measure model (Littell et al. 2006).

Mean separation was determined by the Tukey–Kramer post-hoc test at $P < 0.05$. Soil data were tested to meet assumptions of normality using the Shapiro–Wilk's test from the UNIVARIATE procedure. In cases where the raw data did not conform to a normal distribution, the data were logarithmically transformed prior to analysis.

Soil quality ratings were determined for the 1980s and 2013 data using the Alberta Soil Quality Criteria for the

Plains region (Alberta Agriculture 1987). The 2013 soil ratings were also compared with those for the local soils to examine the degree of soil quality changes in the reclaimed soils relative to the natural sites. The critical properties evaluated for soil quality were pH, EC and SAR. The criteria for rating the topsoil pH range from 6.5 to 7.5 (good), 5.5 to 6.4 or 7.6 to 8.4 (fair), 4.5 to 5.4 or 8.5 to 9.0 (poor), and < 4.5 or > 9.0 (unsuitable). The ratings for topsoil EC are < 2 dS m⁻¹ (good), 2 to 4 dS m⁻¹ (fair), 4 to 8 dS m⁻¹ (poor), and > 8 dS m⁻¹ (unsuitable). The ratings for topsoil SAR are < 4 (good), 4 to 8 (fair), 8 to 12 (poor), and > 12 (unsuitable). The criteria for subsoil are similar to those for topsoil except that the EC boundaries are slightly higher for subsoil.

RESULTS AND DISCUSSION

Vertical Distribution of Soil Chemistry in 2013

The soil chemical properties varied significantly ($P < 0.05$) among soil layers (Table 2). In cases where there was a significant effect of factor, the mean separation for subsoil treatment effect was compared across the row and is indicated by lower case letters. For soil layer effect, lower case letters in parentheses indicate differences in vertical distribution of soil chemistry within the column. The soil pH was neutral to slightly alkaline in most cases. The average pH in the topsoil (0–15 cm) was 6.8, and the value increased with depth to 7.3 in the upper subsoil (15–40 cm) and 7.7 in both the mid subsoil (40–100 cm) and lower subsoil (100–120 cm). The soil layers in SUB-0 (treatment without subsoil) and SUB-25 (25-cm subsoil) generally had a higher pH, with a value of about 8.0 in the lower subsoil (100–120 cm depth). The average pH in SUB-0 and SUB-25 profile was significantly higher ($P < 0.05$) than in SUB-300, but statistically similar to that in other treatments (SUB-50, SUB-100 and SUB-150).

There was a significant main effect of soil layer ($P < 0.05$) on EC whereby the average EC varied as topsoil (0.55 dS m⁻¹) < upper subsoil (2.6 dS m⁻¹) < mid subsoil (5.4 dS m⁻¹) = lower subsoil (4.6 dS m⁻¹) as shown in Table 2. The increase in EC levels with depth indicated that salt accumulation was predominantly by downward leaching of solutes from the upper soil layers. Although the main effect of treatment was not

Table 2. Vertical distribution of soil chemistry and variations among treatments in 2013

Soil property	Effect	P value	Soil layer ^z	Treatment ^y						Average ^x
				SUB-0	SUB-25	SUB-50	SUB-100	SUB-150	SUB-300	
pH	Soil layer (S)	<0.001	Topsoil	6.7	7.1	6.9	6.7	7.0	6.6	6.8(c)
	Treatment (T)	0.002	Upper Subsoil	7.6	7.5	7.4	7.4	7.2	6.9	7.3(b)
	S × T	0.609	Mid Subsoil	7.8	7.9	7.5	7.6	7.5	7.5	7.7(a)
			Lower Subsoil	8.1	7.9	7.7	7.7	7.5	7.5	7.7(a)
			Profile Average	7.6a	7.6a	7.4ab	7.4ab	7.3ab	7.1b	
EC (dS m ⁻¹)	Soil layer (S)	<0.001	Topsoil	0.28(b)	1.2(b)	0.45(b)	0.34(b)	0.52(b)	0.44(b)	0.55(c)
	Treatment (T)	0.606	Upper Subsoil	5.0a(a)	2.5b(b)	2.6b(b)	3.6b(ab)	0.68c(b)	1.2c(b)	2.6(b)
	S × T	0.005	Mid Subsoil	5.7(a)	5.5(a)	5.9(a)	4.9(a)	6.0(a)	4.6(a)	5.4(a)
			Lower Subsoil	3.6b(a)	3.2b(ab)	2.8b(b)	5.2ab(a)	7.1a(a)	5.5ab(a)	4.6(a)
			Profile Average	3.7	3.1	2.9	3.5	3.6	3.0	
TDS (tonnes ha ⁻¹)	Soil layer (S)	<0.001	Topsoil	0.20(c)	0.92(c)	0.40(c)	0.28(b)	0.47(b)	0.40(b)	0.48(c)
	Treatment (T)	0.002	Upper Subsoil	10a(b)	3.6bc(bc)	5.2b(bc)	5.6b(b)	1.0c(b)	1.9c(b)	4.6(bc)
	S × T	0.002	Mid Subsoil	59a(a)	45b(a)	57ab(a)	34c(a)	27c(a)	19c(a)	40(a)
			Lower Subsoil	14(b)	14(b)	12(b)	12(ab)	12(ab)	7.8(ab)	12(b)
			Profile Total	83a	63ab	75ab	52b	40bc	29c	
Na (mg L ⁻¹)	Soil layer (S)	<0.001	Topsoil	33(c)	56(d)	43(d)	23(c)	44(c)	28(c)	38(c)
	Treatment (T)	0.154	Upper Subsoil	847a(b)	261b(c)	293b(c)	387b(b)	101c(b)	99c(b)	331(b)
	S × T	0.037	Mid Subsoil	1210(a)	1010(a)	1127(a)	793(a)	887(a)	660(a)	948(a)
			Lower Subsoil	810(b)	710(b)	583(b)	917(a)	1200(a)	840(a)	843(a)
			Profile Average	725	509	511	530	558	407	
SAR	Soil layer (S)	<0.001	Topsoil	1.3	1.2	1.8	0.97	1.5	0.99	1.3(c)
	Treatment (T)	<0.001	Upper Subsoil	11	4.1	7.1	7.0	3.5	3.2	6.0(b)
	S × T	0.245	Mid Subsoil	24	16	20	15	8.6	7.5	15(a)
			Lower Subsoil	30	20	23	16	11	8.9	18(a)
			Profile Average	17a	11ab	13ab	9.7b	6.2b	5.2b	

^zSampled soil layers, where Topsoil = 0–15 cm; Upper Subsoil = 15–40 cm; Mid Subsoil = 40–100 cm; and Lower Subsoil = 100–120 cm.

^ySoil treatments according to subsoil thickness, where SUB-0 = 0 cm; SUB-25 = 25 cm; SUB-50 = 65 cm; SUB-100 = 135 cm; SUB-150 = 165 cm; and SUB-300 = 335 cm. Mean values with the same lower case letter(s) are not significantly different across the row at $P < 0.05$. Mean values with the same lower case letter(s) in parentheses are not significantly different within the column at $P < 0.05$.

^xAverage data across treatments for each soil layer.

significant there was a statistical soil layer × treatment effect on EC ($P = 0.005$). The significant interaction was due to differences in EC among treatments in the upper and lower subsoils. In the upper subsoil (15–40 cm), the highest EC was in SUB-0, followed by EC levels in SUB-25, SUB-50 and SUB-100, and the lowest EC was in SUB-150 and SUB-300. In the lower subsoil (100–120 cm), there was a greater accumulation of salt in the deeper subsoil treatments (i.e., subsoil depth > 100 cm, which included SUB-100, SUB-150 and SUB-300) compared with the shallower subsoil treatments (i.e., subsoil depth < 65 cm, which included SUB-0, SUB-25 and SUB-50).

The vertical distribution of EC among treatments was influenced by differences in subsoil thickness to the spoil layer. In SUB-0, where the topsoil was placed directly on the spoil without subsoil, the EC in the 15- to 40-cm depth (which was mainly the spoil layer) was significantly higher than in the topsoil (0–15 cm), indicating salt accumulation at the topsoil-spoil interface. This is attributed to restricted leaching as dissolved salts moved into the clay-textured, less permeable spoil layer. In other treatments, where subsoil was included in the profile, there was no statistical difference in EC between the topsoil and the upper subsoil layer (15–40 cm). However, EC levels increased significantly in the mid

subsoil (40–100 cm), particularly in SUB-25 and SUB-50, where the depth of spoil layer from the soil surface was 40 and 80 cm, respectively. There was also a significant increase in EC beneath 40 cm depth in the deeper treatments, particularly in SUB-150 and SUB-300. The 0- to 40-cm layer represents the effective root zone for soils in this region, where the depth of solum in the dominant soil series ranged from 25 to 32 cm (Pedocan 1993). In spite of the greater thickness of subsoils in the SUB-150 and SUB-300 treatments, there was a significant increase in EC in the mid subsoil (beneath 40 cm depth). While this was indicative of salt accumulation with depth, it was probably a reflection of the extent of leaching fronts in the soil profile.

To determine the amounts of salt loading in the soil layers, the mass of TDS was calculated in tonnes per hectare using the measured EC values. There were significant main effects ($P < 0.05$) of soil layer and treatment on TDS, but with a statistical interaction of both factors (Table 2). The significant interaction was due to variability in salt loading among treatments in the upper and mid subsoils. In these soil layers, the greatest amounts of salt were recovered in SUB-0, followed by salt compositions in SUB-25, SUB-50 and SUB-100 and the smallest amounts in SUB-150 and SUB-300. The total mass of salt in the root zone (0–40 cm depth) of

SUB-0 was about 10 t ha^{-1} , twice the amount in SUB-25, SUB-50 and SUB-100 where the average TDS was 5.4 t ha^{-1} , while the average for SUB-150 and SUB-300 was about 2.0 t ha^{-1} . The total salt loading in the entire soil profile (0–120 cm) in SUB-0 was significantly greater than in SUB-100, SUB-150 and SUB-300 by 37, 52 and 65%, respectively ($P < 0.05$). Amounts of total salt in SUB-25 and SUB-50 profiles were statistically similar to that in SUB-0 even though the TDS in SUB-0 was greater than in SUB-25 and SUB-50 by 24 and 10%, respectively. Amounts of TDS also increased with depth in each treatment, but this was due to the differences in soil sampling intervals, since the estimated mass of total salt was a function of depth with varying thicknesses of 0–15, 15–40, 40–100 and 100–120 cm.

The soluble Na data were used to characterize the composition and redistribution of solutes in the soil pore water, while the level of soil sodicity was measured by SAR (the ratio of Na concentration to the sum of calcium and magnesium concentrations). Although there was no main effect of treatment, the concentration of soluble Na was influenced by soil layer $\times$ treatment interaction (Table 2). The interaction of factors was due to differences in Na concentration among treatments in the upper subsoil (15–40 cm). In the upper subsoil, the highest concentration of soluble Na was in SUB-0, followed by Na concentrations in SUB-25, SUB-50 and SUB-100 and the smallest concentrations were in SUB-150 and SUB-300 ($P < 0.05$). The concentration of Na increased significantly with depth in all treatments, indicating downward movement of solute from the topsoil to the upper and mid subsoils. In the shallower treatments (SUB-0, SUB-25 and SUB-50), however, amounts of Na in the mid subsoil (40–100 cm) were higher ($P < 0.05$) than in the lower subsoil (100–120 cm), but the values were statistically similar for the two soil layers in the deeper treatments (SUB-100, SUB-150 and SUB-300). In general, Na concentration varied with depth as topsoil (38 mg L^{-1}) < upper subsoil (331 mg L^{-1}) < mid subsoil (948 mg L^{-1}) = lower subsoil (843 mg L^{-1}).

There were significant main effects ($P < 0.05$) of soil layer and treatment on SAR, but no interaction of factors, indicating a consistent trend in SAR variability among treatments in each soil layer (Table 2). Based on the average for each profile, SUB-0 had the highest SAR, but the value was statistically similar to that in SUB-25 and SUB-50. Nonetheless, there were no statistical differences in SAR among all the treatments containing subsoil materials. The SAR values varied among soil layers as topsoil (1.3) < upper subsoil (6.0) < mid subsoil (15) = lower subsoil (18).

The vertical distribution of soluble Na indicated a substantial accumulation of solutes in the mid subsoil layer (40–100 cm) for the shallower treatments, where subsoil thickness was < 65 cm. While this may be largely due to restricted downward leaching above and within the spoil layer, upward migration of salt possibly contributed to an elevated concentration of Na in the

mid subsoil of the shallower treatments. There was also evidence of salt accumulation within the 40 to 100 cm depth in all treatment plots during the initial 5-yr monitoring (1982–1986), as reported by Leskiw (1989). However, it was not clear whether upward migration contributed significantly to salt accumulation at this depth relative to leaching in the 1980s, as there was no evidence of a shallow water table to promote capillary rise of dissolved salts from the underlying spoil layer (Moran et al. 1987; Leskiw 1989). Due to absence of a shallow water table within the top 3 m depth at the BRSRP site, Leskiw (1989) concluded that the site is a groundwater recharge area and thereby eliminates capillary rise as a mechanism for upward salt movement.

Other mechanisms that may be driving upward migration of salt are diffusion and advective vapour flow. The diffusion process depends on the concentration gradient, which must be high enough to promote upward migration of Na into the overlying soil layer. This may have been possible in the 1980s when the SAR of spoil material was about twice of that in the subsoil (Table 1). However, upward salt migration by diffusion is unlikely to occur by 2013 due to downward movement of Na from the upper soil layer, thereby attenuating the concentration gradient between subsoil and spoil. In a study conducted near the BRSRP site, Macyk et al. (1995) observed a general downward redistribution of salts, and there was no evidence of salt perching at the subsoil–spoil interface except in poorly drained, depressional sites. This is unlike the findings of Kessler et al. (2010) in the Alberta oil sands region. These authors investigated the influence of coversoil thickness on salt redistribution in reclaimed soil covers placed over saline-sodic overburden (also referred to as spoil) during the initial 4 yr after placement. Kessler et al. (2010) reported that diffusion was the dominant mechanism driving upward migration of salt from the spoil into 15 to 20 cm of the cover soils during the early period following placement. Their findings also showed that salt accumulation was largely influenced by downward leaching as soil water percolation was impeded by the low hydraulic conductivity of the sodic overburden. However, model simulation indicated that the solute transport will be dominated by net downward leaching as the concentration gradient decreases with time.

Although it is intuitive to expect water accumulation at the soil–spoil interface due to restricted leaching, there was no evidence of a perched water at the BRSRP site during monitoring in the fall of 1980s and 2013. However, the spoil layer tended to be moist and was generally wetter than the subsoil, but the moisture content was always below field capacity (Leskiw 1989). Perhaps there was a temporary accumulation of water at this depth during peak hydrologic regimes, which could have dried up by evapotranspirative demand by the end of the growing season. This underscores the role of advective vapour flow on upward migration of salt as

the soil profile adjusts to suction and thermal gradients (Parlange et al. 1998; Hillel 2004).

In general, the vertical distribution of pH was concomitant to the trend in salt composition within the soil profile, as the soil salinity levels increased with depth by downward leaching. In spite of the moisture deficit in this region, the vertical distribution of soil parameters reflected the influence of episodic moisture regimes such as growing season storm events and spring snowmelt on solute movement. Based on groundwater recharge characteristic of the BRSRP site, Leskiw (1989) inferred that both saturated and unsaturated transmission of water and salts occurred in the soil profile, thereby promoting solute movement through the soil matrix and preferential flow pathways over time (Beven and Germann 1982; Kung 1990a, b). Given that the site had been under forage since 1987, the long-term influence of root penetration and decayed root channels could result in the formation of an extensive pore network and continuous pore geometry within the root zone. This is also associated with surface fissures, voids and cracks created by cycles of freezing/thawing and swelling/shrinking of the clay loam material, which subsequently enhanced salt movement via the macropores and mesopores (Radulovich et al. 1992; Mitchell and van Genuchten 1993; Flury et al. 1994).

Salt movement could also occur by unsaturated flow via the micropores and capillary pores (Radulovich et al. 1992). Although water and salt movement by preferential flow is expected to bypass the soil matrix, there is a possibility of mass exchange between the two porous media depending on the pore geometry. Various dual-permeability and dual-porosity models have been used to describe dual flow processes in soil (Gerke and van Genuchten 1993; Jarvis 2007). Weiler and Fluhler (2004) also provided evidence of dual flow in macroporous soils using dye tracer experiments under irrigation. The study revealed that both macropore and matrix flow patterns occurred, and there was an interaction between the two systems of flow.

Long-term Changes in Soil Chemistry

Temporal changes in soil chemistry between 1980s and 2013 were presented for each soil layer based on two-way ANOVA of year $\times$ treatment effects. Though there was no significant interaction of year and treatment in all cases, the graphical outputs were used to illustrate the soil chemistry trend over time. To examine the overall changes in soil salinity (EC and TDS) and sodicity (SAR and soluble Na) in the 1980s relative to 2013, the historical soil monitoring data for 1982–1984 were averaged and denoted as Mean-1980s. The baseline soil data were also plotted with the Mean-1980s and 2013 data to illustrate the temporal trends in soil chemistry following soil reconstruction. However, the baseline data were not included in the statistical analysis in order to emphasize the changes in soil parameters between the two monitoring periods (1980s vs. 2013).

The topsoil EC levels were the same across treatments, but the average EC decreased significantly ($P < 0.05$) from 3.1 dS m^{-1} in 1980s to 0.54 dS m^{-1} in 2013 (Fig. 3a). Relative to the baseline, the mean topsoil EC had declined by 33 and 88% in 1980s and 2013, respectively. In the upper subsoil (15–40 cm), the average EC increased slightly to 6.0 dS m^{-1} in 1980s relative to the baseline; indicating accumulation of salt from the topsoil after soil placement (Fig. 3b). Treatment effect on EC in the upper subsoil was noticeable in 2013, whereby the trend in EC among treatments was characterized by three levels of variability. The highest EC was measured in SUB-0 as 5.0 dS m^{-1} , followed by those in the SUB-25, SUB-50 and SUB-100 treatments, where the average EC was 3.0 dS m^{-1} and the lowest EC was in SUB-150 and SUB-300, which was about 1.0 dS m^{-1} ($P < 0.05$).

The pattern of EC distribution among treatments in the upper subsoil (15–40 cm depth) in 2013 reflected the influence of differing subsoil materials on salt fluxes. The upper subsoil in SUB-0 was spoil material unlike in other treatments where the layer was subsoil. As discussed earlier, the spoil material in the SUB-0 plot was restricting downward flushing of salt in the upper subsoil. In contrast, salt movement improved significantly in SUB-25, SUB-50 and SUB-100 but the extent of leaching was statistically similar among these treatments in 2013 while SUB-150 and SUB-300 were most effective in responding to salt leaching in the upper subsoil. Between 1980s and 2013, amounts of EC in the upper subsoil declined by 25% in SUB-0, by about 50% in SUB-25, SUB-50 and SUB-100, and by 83% in SUB-150 and SUB-300.

There was no treatment effect on EC in the mid subsoil (40–100 cm), but salinity levels in 1980s were also higher than the baseline, indicating salt flushing from the upper soil layers (Fig. 3c). The average EC in 2013 declined by 17% from that in the 1980s and was subsequently similar to the baseline. The EC trends in the lower subsoil (100–120 cm) were influenced by subsoil thickness during monitoring in the 1980s and 2013 (Fig. 3d). Amounts of EC recovered in the lower subsoil of the shallower treatments (SUB-0, SUB-25 and SUB-50) were smaller ($P < 0.05$) than for the deeper treatments (SUB-100, SUB-150 and SUB-300). The EC values in the shallower treatments were also lower than in the baseline, while the EC in the deeper treatments was the same as the baseline particularly in 2013. These results indicate a substantial accumulation of salt moved from the upper profile at the lower subsoil of the deeper treatments. As such, the EC trends in the lower subsoil of 2013 monitoring complement those in the upper subsoil. Salt accumulation with depth in the deeper treatments was indicative of the extent of the leaching front in the reclaimed profile, given that the spoil layer was beneath 120 cm depth in the deeper treatments. Nonetheless, there was a significant decline in EC between 1980s and 2013 in most cases in the lower subsoil.

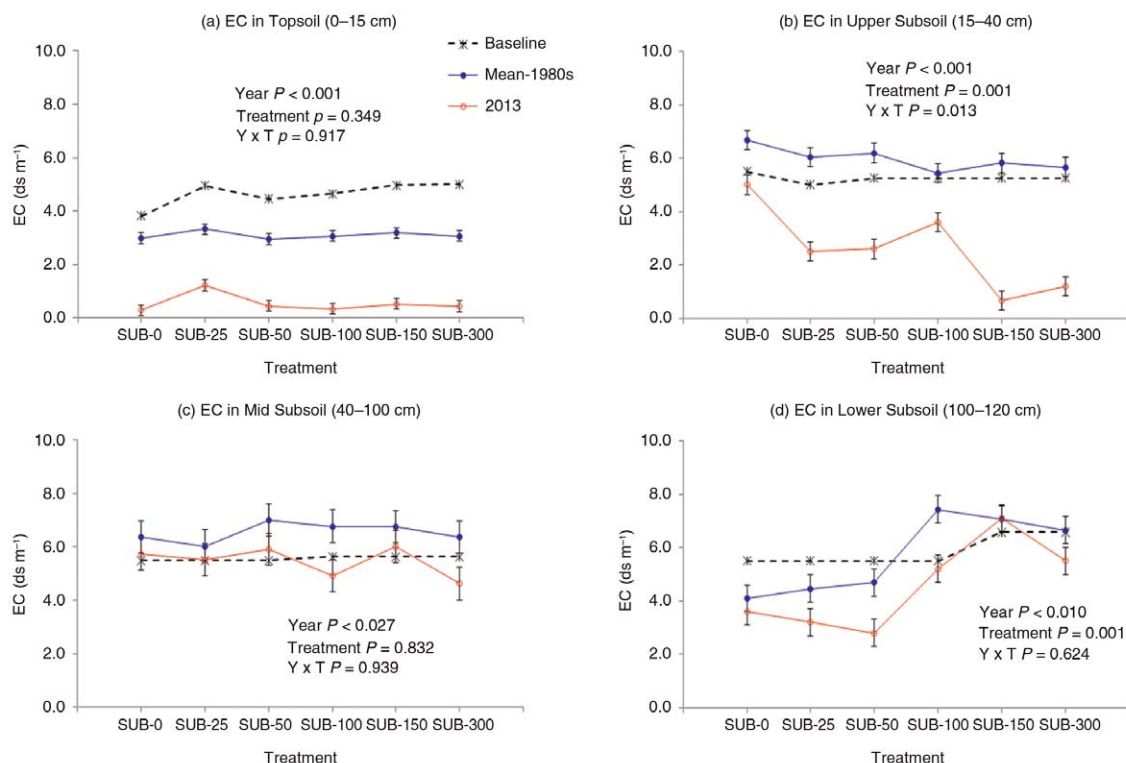


Fig. 3. Temporal variation in EC among treatments in each sampled soil layer: (a) topsoil (0–15 cm); (b) upper subsoil (15–40 cm); (c) mid subsoil (40–100 cm); and (d) lower subsoil (100–120 cm). The vertical bars represent standard errors of the mean.

The average TDS in the topsoil decreased from 4.6 t ha⁻¹ for the baseline to 2.2 t ha⁻¹ in 1980s and subsequently to 0.48 t ha⁻¹ by 2013 (Fig. 4a). The TDS also declined between 1980s and 2013 for all treatments in the upper subsoil (Fig. 4b). The TDS trend in the upper subsoil in 2013 was also characterized by three levels of variability, such that the differences among treatments were SUB-0 (14 t ha⁻¹) > SUB-25 = SUB-50 = SUB-100 (6.7 t ha⁻¹) > SUB-150 = SUB-300 (2.1 t ha⁻¹). There was no main effect of year on TDS in the mid subsoil, but amounts of total salt in the shallower treatments in 2013 were greater than in the deeper treatments (Fig. 4c). The significant year $\times$ treatment effect ($P < 0.05$) on EC in the mid subsoil was due to a decrease in salinity in SUB-0 by 2013, while EC increased over time in SUB-25 and SUB-50, but the value was time-invariant in SUB-100, SUB-150 and SUB-300. There was also a significant year $\times$ treatment effect on TDS in the lower subsoil (Fig. 4d). In the 1980s, the salt loading in the lower subsoil was the same for the shallower treatments with an average TDS of 19 t ha⁻¹, twice the amount in the deeper treatments. By 2013, TDS declined remarkably in the shallower treatments relative to the 1980s, but with a concomitant increase in SUB-100 and SUB-150 treatments, even though the TDS was similar across treatments in 2013.

Sodic levels declined over time in the topsoil and upper subsoil (the two soil layers representing the root

zone) but there was no treatment effect on SAR in both soil layers (Fig. 5a, b). The average SAR in the topsoil was 6.0, 3.3 and 1.3 for the baseline, Mean-1980s and 2013, respectively. The SAR in the upper subsoil increased slightly from 7.7 in the baseline to about 10 in the 1980s, but the value decreased significantly to 6.0 by 2013 ($P < 0.05$). In the upper subsoil, the SAR variability among treatments in 2013 was similar to the trend observed for EC and TDS except that the differences in SAR were not statistically significant. Unlike in the root zone, SAR varied significantly among treatments in the mid and lower subsoils (Fig. 6c, d). In the mid subsoil, the SAR in SUB-0 was higher than in other treatments, where the value was statistically similar in both monitoring periods (Fig. 5c). In the lower subsoil (Fig. 5d), the temporal variation in SAR was more noticeable for the shallower treatments (SUB-0, SUB-25 and SUB-50) as the values diminished in 2013 relative to the 1980s in spite of the SARs in the shallower treatments being higher than in the deeper treatments (SUB-100, SUB-150 and SUB-300).

As observed for EC, the concentration of soluble Na decreased significantly between the 1980s and 2013 in all soil layers (Fig. 6a–d). The trend in Na concentration in the upper subsoil in 2013 also indicated three levels of variability among treatments, as shown for EC and TDS. In the mid subsoil, amounts of soluble Na

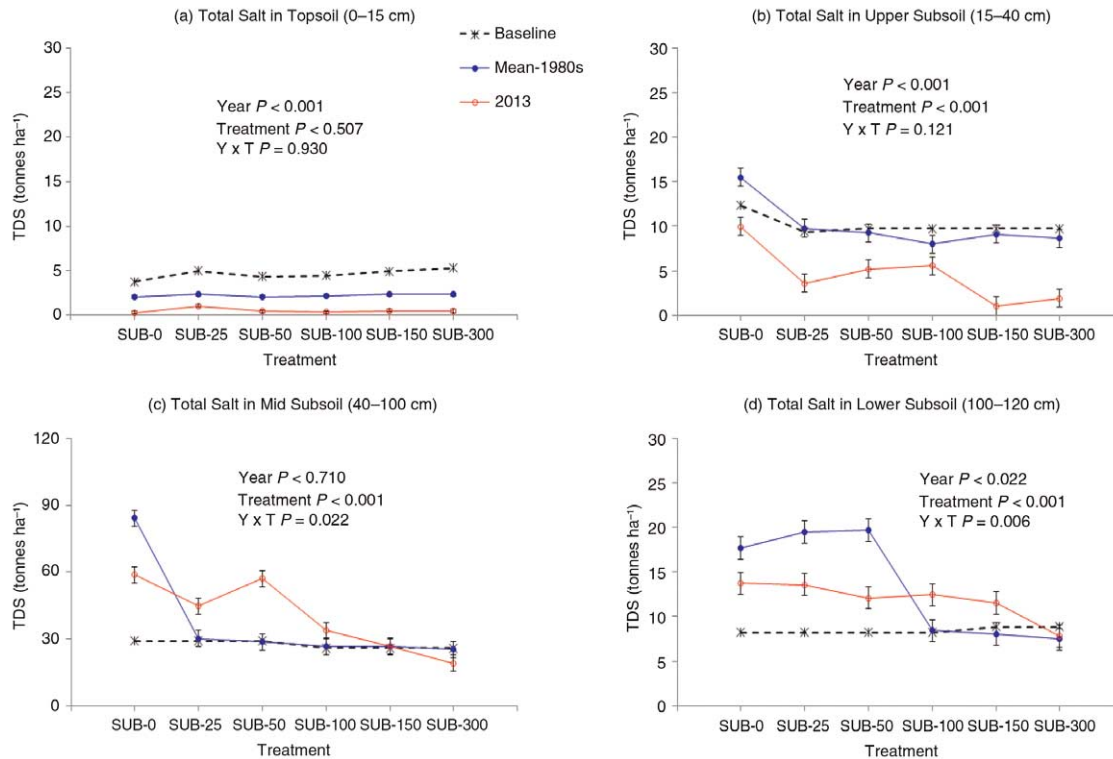


Fig. 4. Temporal variation in total dissolved salts (TDS) among treatments in each sampled soil layer: (a) topsoil (0–15 cm); (b) upper subsoil (15–40 cm); (c) mid subsoil (40–100 cm); and (d) lower subsoil (100–120 cm). Note that the maximum value of TDS on the Y-axis in Fig. 4c is 120 t ha⁻¹. The vertical bars represent standard errors of the mean.

diminished over time in all treatments, but Na concentration varied among treatments relative to the subsoil thickness (Fig. 6c). Unlike in the mid subsoil, there was no treatment effect of Na concentration in the lower subsoil (Fig. 6d).

Soil salinity and sodicity generally decreased over time but the temporal trends depended on the soil layer and subsoil thickness. The root zone (0–40 cm) comprising the topsoil (0–15 cm) and upper subsoil (15–40 cm) was most responsive to temporal decline in soil chemistry. This can be attributed to the high intensity of soil moisture flux and the proximity of the root zone layers to the soil surface. Salt redistribution in the root zone was more influenced by infiltrating water than in the lower soil profile. The influence of pore network in the root zone and downward leaching of dissolved salts was demonstrated by EC and soluble Na as the amounts of these parameters diminished with time in all soil layers. The subsequent increase in EC and Na with depth also corroborated salt flushing by leaching over time. In spite of the moisture deficit at this site, these findings indicated that episodic moisture regimes (growing season rainfall and spring snowmelt) and the presence of fractured pores contributed to substantial vertical redistribution and temporal decline in salt levels in the reclaimed soil profiles.

Salt movement and temporal changes were also influenced by subsoil thickness. Inclusion of subsoil material in the reconstructed soil enhanced salt flushing and temporal changes compared with the topsoil/spoil treatment, but the extent of salt redistribution varied depending on the subsoil thickness. Since the topsoil material was uniformly imposed on all plots, salt composition in the topsoil layer was similar for all treatments, and any differences in concentration of residual salt would be eliminated by intense moisture flux at this depth. In the upper subsoil, however, the downward flushing of salt was impeded in the shallower treatments (subsoil depth < 65 cm) due to the smaller buffer between the topsoil and subsoil–spoil interface compared with the deeper treatments where the spoil layer was beneath 100 cm depth. These results suggest that subsoil thickness influenced the size of the transmission zone and the volume of water supply in the soil profile, which ultimately influenced the extent of salt leaching.

To examine the influence of soil bulk density on salt redistribution, temporal trends in bulk density were determined for each soil layer. There was no statistical effect of treatment on bulk density. Thus, the average bulk density was compared across monitoring years for each soil layer. The data were available for 1983, 1985, 1986, 1987 and 2013 monitoring. In 1983 (2 yr after soil placement), the bulk density in the topsoil (0–15 cm) was

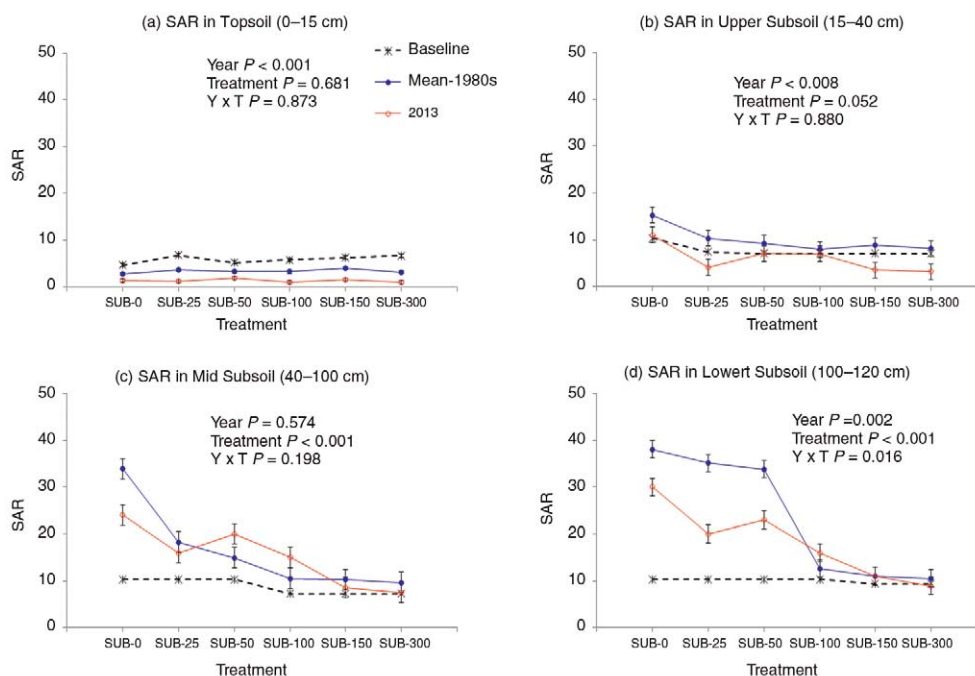


Fig. 5. Temporal variation in SAR among treatments in each sampled soil layer: (a) topsoil (0–15 cm); (b) upper subsoil (15–40 cm); (c) mid subsoil (40–100 cm); and (d) lower subsoil (100–120 cm). The vertical bars represent standard errors of the mean.

1.69 g cm⁻³ (Fig. 7). The bulk density declined significantly to 1.51 g cm⁻³ in 1985 and to 1.40 g cm⁻³ in 1987 ($P < 0.05$). By 2013, the bulk density had declined to 1.21 g cm⁻³, which was 30% lower than that measured in 1983. Unlike in the topsoil, the temporal changes in subsoil densities were rather gradual particularly in the 1980s. In the upper subsoil (15–40 cm), the bulk density was 1.84 g cm⁻³ in 1983, but decreased significantly to 1.54 g cm⁻³ in 2013, while the density for the mid subsoil (40–100 cm) also declined from 1.80 g cm⁻³ in the 1980s to 1.54 g cm⁻³ by 2013 ($P < 0.05$). In the lower subsoil (100–120 cm), however, bulk density remained unchanged over time with a mean of 1.51 g cm⁻³ for the 1980s and 2013.

The bulk density was considered too high in the early 1980s, with implications for compaction, limited pore space and root distribution. In the late 1980s, however, the topsoil bulk density was within the range reported for agricultural soils, but the subsoil densities were still excessive. By 2013, the topsoil and subsoil densities in all treatment plots are comparable to those in agricultural and natural sites. The significant decline in topsoil density is attributed to organic matter cycling and root activities. Other studies conducted at the Battle River region also showed that the bulk density of the reclaimed soil was progressively decreasing over time due to the influence of frost action and root penetration by forage (Moran et al. 1987; Macyk et al. 1995). Graveland et al. (1988) examined the effects of varying subsoil depths on temporal changes in physical properties of the reclaimed soil profile at the Highvale Mine

Permit Area, west of Edmonton, AB. The authors reported a significant decrease in bulk density during the 5-yr study for soil profile reconstructed with 55 cm of subsoil. The reduction in bulk density was attributed to root penetration by alfalfa–brome grass mixed forage (Bailey 2001). It is clear that root activity was the most significant factor reducing the bulk density, since the void space increases as the plant roots penetrate the soil aggregates. This is particularly important if the site has been under long-term forage without cultivation as it was at the BRSRP site. Accumulation of organic matter and decayed plant roots in soils under no-till conditions could promote the soil structure and pore connectivity over time, thereby improving the total porosity, which is inversely related to bulk density.

Overall, salt movement occurred in the treatment plots and was predominantly by downward leaching, possibly through both the soil matrix and preferential flow pathways. According to the observed trend in salt redistribution, leaching is expected to continue over time at the BRSRP site with subsequent salt flushing in the soil profile.

Soil Quality Ratings

To capture the evolution of soil quality following soil reconstruction, suitability ratings of various soil chemical parameters (pH, EC and SAR) were evaluated for the local natural soils and the reclaimed soils. To examine the root zone quality and effectiveness of soil material as plant growth medium, the profile attributes and ratings for the natural soils are emphasized only for

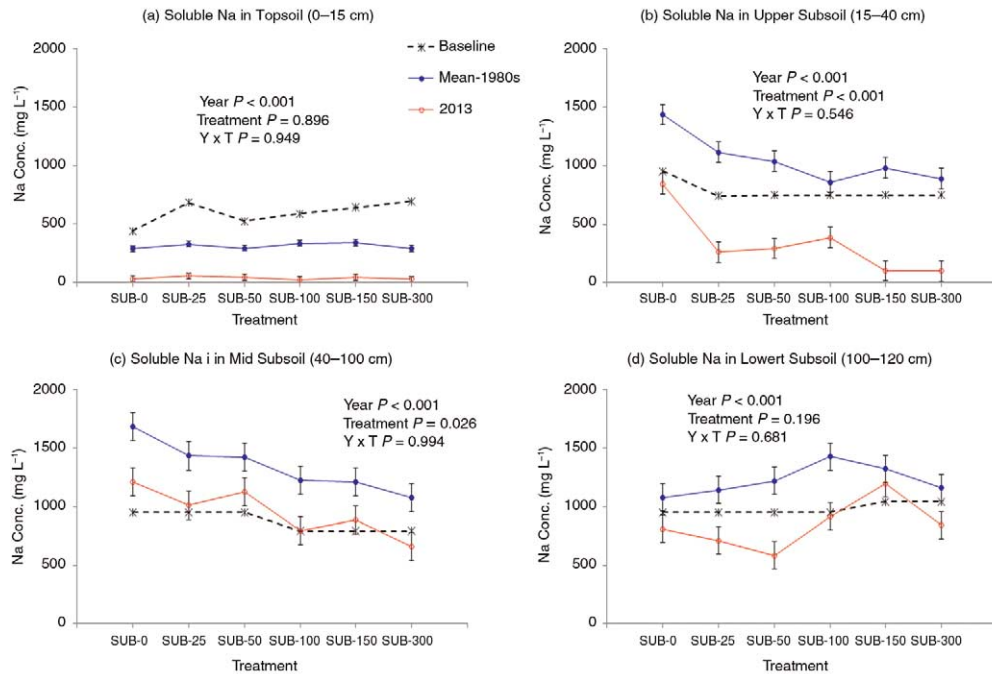


Fig. 6. Temporal variation in concentration of soluble Na among treatments in each sampled soil layer: (a) topsoil (0–15 cm); (b) upper subsoil (15–40 cm); (c) mid subsoil (40–100 cm); and (d) lower subsoil (100–120 cm). The vertical bars represent standard errors of the mean.

the solum (A and B horizons), while the discussion for the reclaimed soils focused on the topsoil (0–15 cm) and upper subsoil (15–40 cm).

For the natural soils, the dominant soil series are the Flagstaff, Halkirk and Torlea Series (Table 3). The Flagstaff Series is a Solonchic Dark Brown Chernozem developed on a moderately fine till (Pedocan 1993). The solum is about 30 cm deep and is composed of a loam Ap horizon (0–12 cm) and clay loam Btnj (12–28 cm). The Ap horizon is rated “Good” for pH, EC and SAR while the B horizon is “Good” for EC but “Fair” for pH and SAR (Alberta Agriculture 1987; Pedocan 1993). The Halkirk and Torlea Series are both classified as Dark Brown Solodized Solonch. The Halkirk Series

is developed on a fine till with a loam-textured Ap (0–12 cm) and clay loam Btn (12–32 cm). The A horizon for the Halkirk Series has a rating of “Good” EC but “Fair” pH and SAR. However, sodicity is the major limitation for the Halkirk B horizon, which has “Unsuitable” SAR while pH and EC are rated “Good” and “Poor”, respectively. The Torlea Series is formed on shallow fine softrock with a clay loam Ah (0–5 cm) and clay-textured Btn (5–25 cm). The A horizon pH and EC are both rated “Good” but “Poor” for SAR. Unlike for other soils, the B horizon of the Torlea Series has the highest limitation for soil chemistry as the pH, EC and SAR are rated “Fair”, “Fair” and “Unsuitable”, respectively. As indicated by the solum depth, the effective root zone of natural soils at this site is about 30 cm and the greatest limitation is sodicity.

In the reconstructed soil profiles, the pH of soil layers within the root zone (0–40 cm) was rated “Good” following soil placement in 1981 and by 2013, but the rating degraded to “Fair” in the upper subsoil (15–40 cm) of SUB-0 treatment due to increase in pH in 2013 (Table 4). In the 1980s, the EC rating for all treatments was “Fair” for the topsoil and “Poor” for the upper subsoil but both soil layers improved to “Good” by 2013 except for the upper subsoil of SUB-0 and SUB-100 treatments, which only improved to “Fair”. The topsoil SAR was “Good” during the 1980s and continued through to 2013. In the upper subsoil, however, SAR was rated “Poor” in most cases in the 1980s. By 2013, the SAR remained “Poor” in SUB-0 and “Fair” in SUB-100,

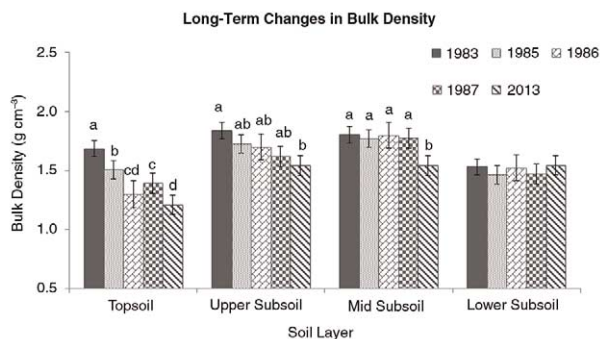


Fig. 7. Temporal variations in soil bulk density in the sampled soil layers. Bars with the same letter(s) are not significantly different at $P < 0.05$.

Table 3. Profile attributes and soil quality ratings for the local natural soils (Pedocan 1993)

Soil series	Parent material	Horizon ^z	Depth (cm)	Texture	pH	EC (dS m ⁻¹)	SAR	Sat%
				Soil quality ratings ^y				
Flagstaff (Solonetzic Dark Brown Chernozem)	Moderately fine till	Ap	0–12	Loam G	6.9 G	1.3 G	1.7 G	40 G
		Bt _{nj}	12–28	Clay Loam F	7.6 F	1.2 G	6.2 F	51 G
		Csk ₁	28–80	Clay Loam F	8.1 F	7.1 P	8.8 P	50 G
		Csk ₂	80–180	Clay Loam F	8.1 F	10.9 U	15.6 U	43 G
Halkirk (Dark Brown Solodized Solonetz)	Moderately fine till	Ap	0–12	Loam G	5.7 F	0.4 G	7.7 F	44 G
		Bt _n	12–32	Clay Loam F	7.3 G	6.3 P	30.5 U	71 F
		Csk	32–180	Clay Loam F	8.0 F	11.4 U	23.2 U	69 F
Torlea (Dark Brown Solodized Solonetz)	Moderately fine softrock	Ah	0–5	Clay Loam F	7.5 G	1.3 G	9.1 P	63 F
		Bt _n	5–25	Clay P	8.3 F	3.3 F	29.5 U	160 U
		Csk	25–180	Clay P	8.2 F	13.2 U	35.5 U	136 U

^zHorizon designation according to the Canadian System of Soil Classification.

^ySoil quality ratings according to the Alberta Soil Quality Criteria for the Plains region (Alberta Agriculture 1987), where G = Good; F = Fair; P = Poor; and U = Unsuitable.

but the soil quality improved from “Poor” to “Fair” in SUB-25 and SUB-50 and from “Poor” to “Good” in SUB-150 and SUB-300, where the subsoil was thicker. Sodicity is also the greatest limitation to root zone quality in the reclaimed soil. This was particularly critical in SUB-0 (treatment without subsoil) but soil quality ratings for pH, EC and SAR in SUB-25 (treatment with 25 cm subsoil) were similar to those in treatments with deeper subsoil in most cases. Soil parameter ratings for the SUB-25 treatment demonstrate that the addition of

25 cm of subsoil to 15 cm topsoil improved the root zone quality and was sufficient to create a suitable coversoil for capping the spoil layer.

While solum depth for the natural soils is 30 cm, the root zone of the reclaimed soil profile was 40 cm deep (i.e., the combination of topsoil and upper subsoil). The root zone configuration in SUB-0 treatment (i.e., topsoil/spoil) was equivalent to the solum of the Torlea Series, which is a typical Solonetzic soil with shallow bedrock. On the other hand, the root zone layers of SUB-25

Table 4. Soil quality ratings of pH, EC and SAR for the reclaimed soils in 1980s and 2013

Soil Property	Soil layer ^z	Year	Treatments ^y					
			SUB-0	SUB-25	SUB-50	SUB-100	SUB-150	SUB-300
pH	Topsoil	1980s ^x	7.0 G	7.2 G	7.0 G	7.1 G	7.4 G	7.5 G
		2013	6.7 G	7.1 G	6.9 G	6.7 G	7.0 G	6.6 G
	Upper Subsoil	1980s ^x	7.3 G	7.2 G	7.3 G	7.3 G	7.3 G	7.3 G
		2013	7.6 F	7.5 G	7.4 G	7.4 G	7.2 G	6.9 G
	Mid Subsoil	1980s ^x	7.3 G	7.3 G	7.3 G	7.5 G	7.5 G	7.5 G
		2013	7.8 F	7.9 F	7.5 G	7.6 F	7.5 G	7.5 G
	Lower Subsoil	1980s ^x	7.3 G	7.3 G	7.3 G	7.3 G	7.7 F	7.7 F
		2013	8.1 F	7.9 F	7.7 F	7.7 F	7.5 G	7.5 G
EC (dS m ⁻¹)	Topsoil	1980s	2.9 F	3.3 F	3.0 F	3.1 F	3.2 F	3.1 F
		2013	0.28 G	1.2 G	0.45 G	0.34 G	0.52 G	0.44 G
	Upper Subsoil	1980s	6.7 P	6.1 P	6.2 P	5.4 P	5.8 P	5.7 P
		2013	5.0 F	2.5 G	2.6 G	3.6 F	0.68 G	1.2 G
	Mid Subsoil	1980s	6.4 P	6.0 P	7.0 P	6.8 P	6.8 P	6.4 P
		2013	5.7 P	5.5 P	5.9 P	4.9 F	6.0 P	4.6 F
	Lower Subsoil	1980s	4.1 F	4.5 F	4.7 F	7.4 P	7.1 P	6.7 P
		2013	3.6 F	3.2 F	2.8 G	5.2 F	7.1 P	5.5 P
SAR	Topsoil	1980s	2.8 G	3.6 G	3.3 G	3.2 G	4.0 G	3.2 G
		2013	1.3 G	1.2 G	1.8 G	0.97 G	1.5 G	0.99 G
	Upper Subsoil	1980s	15 P	10 P	9.3 P	8.0 F	8.8 P	8.1 P
		2013	11 P	4.1 F	7.1 F	7.0 F	3.5 G	3.2 G
	Mid Subsoil	1980s	34 U	18 P	15 P	11 P	10 P	9.6 P
		2013	24 U	16 P	20 U	15 P	8.6 P	7.5 F
	Lower Subsoil	1980s	38 U	35 U	34 U	13 P	11 P	11 P
		2013	30 U	20 U	23 U	16 P	11 P	8.9 P

^zSampled soil layers, where Topsoil = 0–15 cm; Upper Subsoil = 15–40 cm; Mid Subsoil = 40–100 cm; and Lower Subsoil = 100–120 cm.

^ySoil treatments according to subsoil thickness, where SUB-0 = 0 cm; SUB-25 = 25 cm; SUB-50 = 65 cm; SUB-100 = 135 cm; SUB-150 = 165 cm; and SUB-300 = 335 cm. Soil quality ratings according to the Alberta Soil Quality Criteria for the Plains region, where G = Good; F = Fair; P = Poor; and U = Unsuitable.

^xSoil pH from the baseline measurements in 1981 due to lack of pH data in the 1980s monitoring.

treatment were comparable to those of the Flagstaff (Chernozem) and Halkirk Series (Solonetz with deeper horizon). As such, SUB-0 and SUB-25 are the two subsoil treatments with root zone configurations similar to those in the natural soils. By 2013, the root zone quality in SUB-0 plot was better than in the Torlea Series, particularly for SAR, which was rated “Good” in the reclaimed topsoil and “Poor” in the upper subsoil compared with “Poor” and “Unsuitable” ratings for the Torlea Ah and Bnt, respectively. For the SUB-25 treatment, the root zone chemistry was similar to that in the solum of the Flagstaff Series, but ratings for pH, EC and SAR in the reclaimed soil layers were much better than in the Halkirk Series. These results indicate that the root zone quality of the reconstructed soil with and without 25 cm subsoil were comparable and/or superior to that in the counterpart natural soils. This also confirmed that 25 cm is an acceptable depth of subsoil for reconstructing soils on sodic mine spoil in this region.

SUMMARY AND CONCLUSIONS

The fundamental challenges limiting reclamation of post-disturbed coal sites in east-central Alberta were attributed mainly to elevated soil sodicity and salinity. To investigate the most effective soil reconstruction method for mitigating this limitation, subsoil thickness was varied as 0, 25, 65, 135, 165 and 335 cm overlain by 15 cm topsoil. The results showed evidence of salt leaching from the upper soil layers with improved soil quality over the past 31 yr (1982 to 2013). The treatment without subsoil was least responsive to leaching, but the extent of salt movement in the root zone (0–40 cm) of treatment with 25 cm subsoil was comparable with that with 65 cm and 135 cm subsoil thickness, while the greatest leaching occurred in treatments with 165 cm and 335 cm subsoil. There was a substantial accumulation of Na at the 40- to 100-cm depth for treatments with subsoil thickness <65 cm. This is largely attributed to restricted downward leaching above and within the spoil layer, but with minor contribution by upward salt migration from the spoil. Regardless of treatment effect on salt leaching, the root zone salinity and sodicity in 2013 were better than in the 1980s. The observed trend in salt redistribution is expected to continue over time with subsequent salt flushing in the soil profile.

Relative to the native soils in 2013, the root zone soil quality ratings for EC and SAR in the topsoil/spoil treatment were better than in the shallow-bedrock Solonetzic soil, while the topsoil/25-cm subsoil treatment was comparable with the Chernozemic soil, but better than the fine till Solonetzic soil. These findings demonstrate that surface-mined coal areas in the Alberta Plains region can be successfully reclaimed to support agricultural operations. The study indicated that a minimum of 25 cm subsoil with 15 cm of topsoil is required for mitigating salinity and sodicity in these reconstructed soils. Other related experiments at the site are investigating the potential utilization of various soil amendments for land reclamation.

The combination of suitable amendment and recommended depths of topsoil and subsoil is expected to enhance the agricultural capability of post-disturbed lands in this region. We also recommend long-term monitoring of soil quality at 5- or 10-yr intervals as this would be more informative than the short period of monitoring following soil reconstruction. Model simulation may also be conducted to support field measurements and to predict water and salt fluxes under a given climate condition or for a particular timeframe.

ACKNOWLEDGEMENTS

Many thanks to Sherritt Coal (Paintearth Mine) for funding the 2013 soil monitoring project. We also acknowledge Maxxam Analytics Inc. for conducting laboratory analyses for this research at reduced costs. Thanks to various personnel with Paragon Soil and Environmental for field sampling and technical support.

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